Transformers and Attention

Synthetic educational document created for RAG development and evaluation practice. It is intended as a controlled development dataset, not as an authoritative source.

1. Transformer Overview

The Transformer is a neural network architecture introduced for sequence processing. Unlike recurrent architectures, a Transformer can process sequence positions in parallel during training. Its central mechanism is attention, which allows the model to relate one token to other tokens in the sequence.

2. Attention

Attention calculates how strongly one token should consider other tokens. In scaled dot-product attention, queries, keys, and values are used to compute attention weights. A simplified expression is Attention(Q,K,V) = softmax(QK^T / sqrt(dk))V. The attention weights determine how information from the value vectors is combined.

3. Self-Attention

In self-attention, queries, keys, and values are derived from the same input sequence. This allows each token to incorporate information from other tokens. For example, when processing a sentence containing a pronoun, self-attention can help the model relate that pronoun to relevant words elsewhere in the sentence.

4. Multi-Head Attention

Multi-head attention runs several attention operations with different learned projections. Each head can learn to focus on different relationships or aspects of the input. The outputs of the heads are combined and projected to form the final multi-head attention output.

5. Positional Information

Because self-attention does not inherently process tokens sequentially, Transformer models need a way to represent token position. Positional encodings or learned positional embeddings provide information about where tokens occur in a sequence.

6. Encoder and Decoder

The original Transformer architecture contains encoder and decoder stacks. The encoder processes an input sequence into contextual representations. The decoder generates an output sequence while using causal masking and attention to relevant encoder representations. Encoder-only architectures are commonly used for understanding tasks, while decoder-only architectures are widely used for autoregressive text generation.

7. Why Transformers Matter

Transformers became highly influential because they scale effectively with data and computation and can model long-range relationships using attention. They form the foundation of many modern language models and are also used in vision, speech, multimodal systems, and retrieval-related applications.